How-To Library

Informatica

Importing and Exporting Assets Between Amazon Redshift and Enterprise Data Preparation

© Copyright Informatica LLC 2018, 2022. Informatica, the Informatica logo, and Big Data Management are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

You can import data assets from Amazon Redshift to your data lake, and you can export data assets from your data lake to Amazon Redshift. Review this article to prepare Enterprise Data Preparation to import and export data assets.

Supported Versions

- Enterprise Data Preparation 10.2 x
- Enterprise Data Preparation 10.4.0

Table of Contents

Overview. 2
Download the Amazon Redshift JDBC Driver. 2
Create JDBC Connections to Access Amazon Redshift. 3
Add Custom Properties to the Enterprise Data Preparation Service. 4

Overview

You can import data from a database in your Amazon Redshift cluster into your data lake. You can also export data from your data lake to an Amazon Redshift database.

After you import a data asset from Amazon Redshift, you can add the imported data to a project in Enterprise Data Preparation as worksheet. After you prepare an asset and export the data to Amazon Redshift, you can perform further operations on the data in Amazon Redshift.

To prepare to import and export data assets between Amazon Redshift and Enterprise Data Preparation, complete the following steps:

1. Download the Amazon Redshift JDBC Driver, and then copy the driver to the machine where the Enterprise Data Preparation Service runs. You must also copy the driver to the machine where the Data Integration Service associated with Enterprise Data Preparation runs.
2. Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.
3. Add custom properties required to import and export data to the Enterprise Data Preparation Service.

Download the Amazon Redshift JDBC Driver

Download the JDBC 4.1 compatible version of the Amazon Redshift JDBC Driver, and then copy the driver to the machine where the Data Integration Service associated with Enterprise Data Preparation runs.

When you import or export data, the Data Integration Service uses Sqoop to connect to the cluster, and then executes a Sqoop job on the cluster to import or export the data. Sqoop uses the driver to connect to the Amazon Redshift database when the import or export job runs on the cluster.

1. Download the `RedshiftJDBC41-1.1.10.1010.jar` file, which contains the JDBC 4.1 compatible version of the Amazon Redshift JDBC Driver.

   You can download the .jar file from the following link:

   https://s3.amazonaws.com/redshift-downloads/drivers/RedshiftJDBC41-1.1.10.1010.jar

2. Copy the .jar file to the following directory on the machine where the Data Integration Service runs:

   `<Informatica installation directory>/externaljdbcjars`

Create JDBC Connections to Access Amazon Redshift

Create one or more JDBC connections to import and export data assets. You might create different connections for import and export operations.

You include the arguments Sqoop uses to connect to the database in your Amazon Redshift cluster in each JDBC connection.

1. In the Administrator tool, click the **Manage** tab > **Connections** view.
2. In the Domain Navigator, click **Actions** > **New** > **Connection**.

   The New Connection dialog box appears.
3. In the New Connection dialog box, select **Database** > **JDBC**.
4. Click **OK**.

   The **New Connection** wizard appears.
5. Enter the JDBC connection properties required to connect to a database in your Amazon Redshift cluster.

   The following table describes the connection properties you enter:

Property	Description
Name	Name of the JDBC connection. The name is not case sensitive, and must be unique within the domain. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: `~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /`
ID	String that the Informatica application services use to identify the connection. The string is not case sensitive. The string must be 255 characters or less, and must be unique in the domain.
Description	Description of the connection. The description cannot exceed 765 characters.
User Name	Database user name.
Password	Password for the database user name.
JDBC Driver Class Name	Name of the Amazon Redshift JDBC driver class. Enter the following value: `com.amazon.redshift.jdbc41.Driver`

Connection String	Connection string to connect to the database in your Amazon Redshift cluster. Use the following format for the connection string:
	`jdbc:redshift://<Redshift cluster endpoint>:<Redshift cluster port>/<database name>`
	The following example shows a sample connection string:
	`jdbc:redshift://edlredshift.mycluster.us-west-2.redshift.amazonaws.com:5439/mydatabase`

Property	Description
Environment SQL	Optional. SQL commands to set the database environment when you connect to the database. The Data Integration Service executes the connection environment SQL each time it connects to the database.
Transaction SQL	Optional. SQL commands to set the database environment when you connect to the database. The Data Integration Service executes the SQL at the beginning of each transaction.
SQL Identifier Character	Select " " from the drop down list to connect to the database from a Hortonworks HDP or from a Cloudera CDH cluster.
Support Mixed-case Identifiers	Select **Yes** from the drop down list.
Use Sqoop Connector	Select **Sqoop v1.x** to enable Sqoop connectivity.

Sqoop Arguments	Sqoop arguments required to connect to the database.
	Specify the connection string to connect to the database in your Amazon Redshift cluster in the `--connect` argument. Specify the Amazon Redshift JDBC driver Sqoop uses in the `--driver` argument.
	Use the following format for the argument string:
	`--connect jdbc:redshift://jdbc:redshift://<Redshift cluster endpoint>:<Redshift cluster port>/<database name> --driver <JDBC driver class name>`
	The following example shows a sample argument string:
	`--connect jdbc:redshift://edlredshift.cunktrspazuv.us-west-2.redshift.amazonaws.com:5439/mydatabase --driver com.amazon.redshift.jdbc41.Driver`

6. When you finish entering connection properties, click **Test Connection** to test the connection.
7. Click **Finish**.

Add Custom Properties to the Enterprise Data Preparation Service

Use the Administrator tool to add the SchemaNameSameCaseConnections and the sqoopSchemaLowerCase custom properties to the Enterprise Data Preparation Service.

Set the value of the SchemaNameSameCaseConnections custom property to the list of connections that Enterprise Data Preparation uses to import and export data assets. The custom property forces Enterprise Data Preparation to retain the uppercase and lowercase letters used in the name of a schema to which you import or export data.

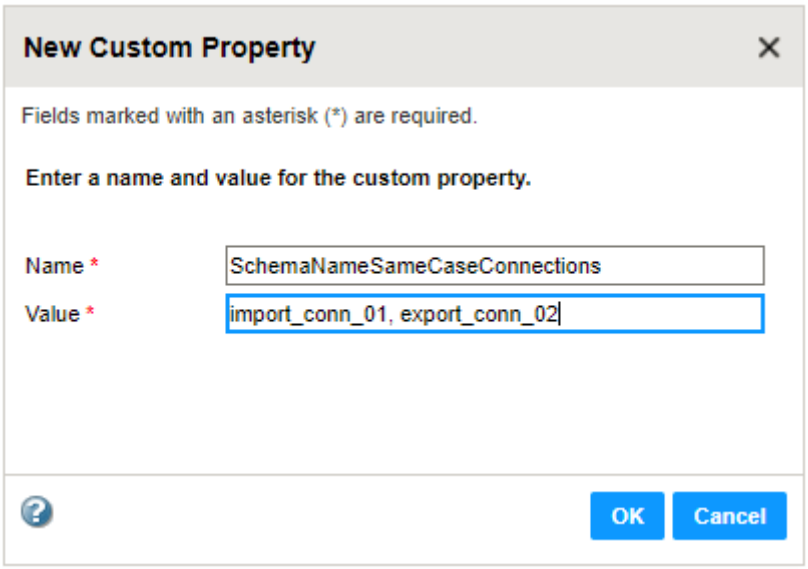
By default, Enterprise Data Preparation converts schema names to uppercase. Schema names used in import and export operations must include the same uppercase and lowercase letters as the schema names in an Amazon Redshift database.

Set the value of the sqoopSchemaLowerCase custom property to the list of connections Enterprise Data Preparation uses to export data assets. The property forces Enterprise Data Preparation to convert the name of the table to which you export data to lowercase. Amazon Redshift uses lowercase table names.

1. In the Administrator tool, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the Enterprise Data Preparation Service.
3. Click the **Properties** view.
4. Click the pencil icon in the Custom Options section.

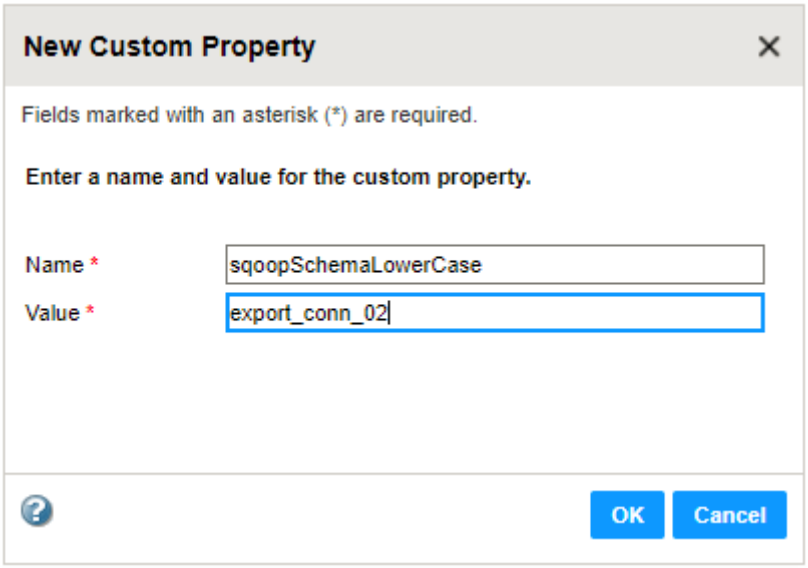
5. Click **New** in the Edit Custom Options dialog box.
6. Enter SchemaNameSameCaseConnections as the custom property name.
7. Enter the name of each JDBC connection Enterprise Data Preparation uses to import and export data as the custom property value. If you enter multiple connection names, separate each connection name with a comma.

   The following image shows the **New Custom Property** dialog box:



8. Click **OK**.
9. Open the Edit Custom Options dialog box again.
10. Enter sqoopSchemaLowerCase as the custom property name.
11. Enter the name of each JDBC connection Enterprise Data Preparation uses to data as the custom property value. If you enter multiple connection names, separate each connection name with a comma.

    The following image shows the **New Custom Property** dialog box:



12. Click **OK**.

You can now import and export data assets between Enterprise Data Preparation and a database in your Amazon Redshift cluster.

For more information about importing and exporting data assets, see the "Import, Export, and Upload Data" chapter in the Enterprise Data Lake User Guide.

Authors

Ram Singh

Aravind Kumar Arunachalam

Manoj Rajan

Swathi N.R.

Dan Hynes